

Ethan Tubbe

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Skills

Languages: Python, JavaScript, Bash, SQL, HTML/CSS

Frameworks/Libraries: Django, Arduino, FastAPI

Databases: PostgreSQL

Tools & Platforms: Git, Docker, Docker Compose, Linux, Tailscale

Projects

Homelab Observability Platform | *Python, PostgreSQL, Docker, SQL, psycpg3* github.com/ethan-pt/logging

- Architected a PostgreSQL schema separating monitoring, logging, event, and security domains, with indexed query paths optimized for time-series and service-level analysis
- Built a containerized data collection pipeline using Docker Compose, enabling independent collectors to register services, emit telemetry, and operate with minimal coupling
- Developed a fault-tolerant collector framework in Python with automatic reconnection, exponential backoff/retry logic, and buffered writes to ensure resilience under database or network failures
- Implemented service-level metadata tracking to associate all telemetry with specific services, enabling scalable multi-service observability

STASH: IoT Catnip Dispenser | *Python, C++ (Arduino), FastAPI, ESP32, HTTP, JSON* github.com/ethan-pt/STASH

- Designed and implemented firmware for an ESP32 to accept user input, manage hardware peripherals, and communicate with a backend service over HTTP
- Developed a FastAPI-based backend service to handle authenticated device requests, track usage metrics, and enforce control logic for physical actions
- Implemented robust network handling to compensate for semi-functional hardware, including connection retries, timeout handling, and automatic recovery via device restart
- Designed a request/response protocol using JSON serialization to coordinate real-time decision-making between device and server

MCMAN: Game Server Deployment Tool | *Python, Docker, Docker Compose, YAML, REST* github.com/ethan-pt/mcman

- Designed and implemented a Python-based CLI for spinning up Minecraft servers with configurable parameters including versioning, resource allocation, and networking
- Programmatically generated Docker Compose configurations, enabling rapid, repeatable deployment of isolated server instances
- Abstracted multiple server types (vanilla, CurseForge, Modrinth, FTB) into a unified interface, prioritizing flexibility and simplicity
- Automated deployment workflows via subprocess orchestration, reducing server uptime from manual configuration to a single command

Reading Tracker | *Python, Django, PostgreSQL, GCP* github.com/ethan-pt/reading_tracker

- Deployed a containerized production application to Google Cloud Platform on Cloud Run with Cloud SQL for managed PostgreSQL and Secret Manager for credential management
- Integrated the Google Books API to support flexible search queries to dynamically display book metadata
- Implemented server-side rendering and request handling to manage user interactions and application state
- Designed and implemented relational data models for users, books, and reading state using Django's ORM

Additional Experience

Line Cook, Various Restaurants – Olympia, WA 2019 – Present

- Eight years of professional experience managing complex, fast-paced workflows through effective teamwork and a commitment to mentoring others. Expert at maintaining high-quality output and operational uptime in mission-critical environments where reliability and clear communication are paramount

Education

Tumwater High School – High School Diploma 2013 – 2017